

Jason Stumpff, Ph.D.
CURRICULUM VITAE

Position: Professor
Department of Molecular Physiology and Biophysics

Address: HSRF 106
Department of Molecular Physiology and Biophysics
The Robert Larner, M.D. College of Medicine
University of Vermont
Burlington, VT 05405
Voice: (802) 656-7849
email: jstumpff@med.uvm.edu
website: <http://physioweb.uvm.edu/stumpff-lab/>

EDUCATION

<u>Year</u>	<u>Institution</u>	<u>Degree</u>	<u>Area of Degree, Accolades</u>
2004	University of Colorado, Boulder	Ph.D.	Molecular, Cellular, and Developmental Biology
1998	Eckerd College	B.S.	Biology Major, Chemistry Minor, with Highest Honors

LICENSES, CERTIFICATION

<u>Years</u>	<u>License/Certification</u>
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FACULTY POSITIONS HELD

<u>Years</u>	<u>Institution</u>	<u>Academic Title</u>	<u>Department</u>
2024-present	University of Vermont	Professor	Molecular Physiology and Biophysics
2019- 2024	University of Vermont	Associate Professor	Molecular Physiology and Biophysics
2011-2019	University of Vermont	Assistant Professor	Molecular Physiology and Biophysics

OTHER POSITIONS AND MAJOR ADMINISTRATIVE POSITIONS HELD

<u>Years</u>	<u>Location/Program Name</u>	<u>Role</u>
2023-present	University of Vermont Cancer Center	Program Leader
2005-2011	University of Washington, Seattle	Postdoctoral Fellow
1998-2004	University of Colorado, Boulder	Graduate Research Assistant

HONORS AND AWARDS

<u>Year</u>	<u>Name of Award</u>
1997	Summer Undergraduate Research Exp Fellowship, Univ. of North Carolina, Chapel Hill
1998	B.S. with Highest Honors in Biology, Eckerd College
2000-2002	NIH Signal Transduction and Cell Cycle Training Grant, Univ of CO
2001	Beverly Sears Award, Univ of CO
2006-2008	NIH Ruth L. Kirschstein National Research Service Award, Postdoctoral Fellowship
2007	First Place Winner, American Society for Cell Biology Imaging Contest
2010	Honorable Mention, American Society for Cell Biology Imaging Contest
2010	Special Fellow Award, Leukemia and Lymphoma Society
2014	Basil O' Connor Starter Scholar Research Award, March of Dimes
2016	Career Catalyst Research Award, Susan G. Komen
2016	Rising Star New Investigator Award, UVM Larner College of Medicine
2021	Mid-Career Investigator Award, UVM Larner College of Medicine
2024	Research Mentor Award, UVM Larner College of Medicine
2026	Juckett Faculty Scholar Award, Juckett Foundation

KEYWORDS/AREAS OF INTEREST

Cell division, mitosis, mitotic spindle, chromosome segregation, cytoskeleton, kinesin, microtubules, molecular motor proteins, aneuploidy, chromosome instability, cancer, triple negative breast cancer

SUMMARY OF PROFESSIONAL ACTIVITIES- OVERALL

I was recruited to the University of Vermont as a tenure-track Assistant Professor in the Molecular Physiology and Biophysics Department of the Larner College of Medicine (LCOM) through a national search. My appointment began on July 1st, 2011, and I was promoted to Associate Professor in July 2019 and to Professor in July 2024. My professional activities since becoming a faculty member at UVM have targeted a split of 60% research, 25% teaching, and 15% service.

Research: My research focuses on understanding the molecular mechanisms that control the cytoskeleton and chromosome organization during cell division and cell differentiation. More specifically, our lab utilizes a combination of quantitative cell imaging and single molecule biophysics to investigate the functions and regulation of kinesin-family motor proteins during mitosis and neurogenesis. This work has been funded by nationally competitive grants from private and government sources since my arrival at UVM in 2011. These include awards from the Leukemia and Lymphoma Society, the March of Dimes, Susan G. Komen, and the National Institutes of Health, as well as scientific research agreements with two pharmaceutical companies to pursue drug development for a promising cancer target we identified. In addition, I have been awarded several pilot grants from the University of Vermont Cancer Center/Lake Champlain Cancer Research Organization and the American Cancer Society to support new research initiatives. Our work has been published in top journals in the field of cell and developmental biology as primary research papers, reviews, and book chapters.

Teaching: My formal teaching at UVM is primarily directed towards graduate and medical students. In 2013, I developed a new graduate-level cell biology course (MPBP 310), which I have directed each time it has been offered. I am also a regular instructor in the Foundations of Clinical Science course for first-year medical students and two core cell biology courses for the CMB PhD program and Medical Masters program (CLBI 301, CLBI 401). I have served as the primary research advisor for 2 postdoctoral fellows, 3 medical fellows, 7 Ph.D. students, and 27 undergraduate students. These focused teaching experiences in the research laboratory setting allow me to guide the career paths of young investigators and are particularly rewarding. A number of my mentees have been awarded competitive research fellowships of their own from NIH, NSF, and NASA, and all have moved on to their positions of choice in research and medicine.

Service: I have provided service at the department, college, and university levels at UVM, as well as to research focused organizations outside of UVM. I am a part of the Senior Leadership Team for the UVM Cancer Center, serving as Program Leader of the Cancer Host and Environment Program. I chaired the UVM Cancer Center Pilot Grant Review Committee for 2 years. I served as chair of the Admissions and Recruitment Committee for the Cellular, Molecular, and Biomedical Sciences graduate program for 4 years and was a member of the committee for 9 years. I have also served on 8 faculty search committees, 2 as chair, and have been a dissertation committee member or chair for 30 graduate students at UVM and 7 at other institutions. Outside of UVM, I was a member of the Finance and Audit Committee for the American Society of Cell Biology for 6 years, serve regularly as an ad hoc grant reviewer for the NIH and other funding agencies, was on the scientific organizing committee for an international meeting in 2022-2023, and frequently review manuscripts for leading journals in my field. I have also participated in or led several educational outreach efforts for the local community.

Inclusive Excellence: I have actively initiated and supported efforts to improve inclusive excellence within STEM research and education. This includes implementing application review policies for the CMB graduate program to enhance recruitment of PhD students from under-represented groups and working with LCOM leadership to establish a scholarship program that supports research-related costs for under-represented students in CMB. I have participated as a mentor in programs that match under-represented undergraduates with advisors at UVM for full-time summer research projects. I have served as a mentor in the LCOM Mentoring Program for Medical Students Under-Represented in Medicine. Finally, my lab has participated in numerous outreach activities that introduce research to high school students in rural areas.

SUMMARY OF ACCOMPLISHMENTS

- Internationally recognized research program focused on understanding how chromosomes are moved and organized during cell division from the single molecule to whole organism level. This work has led to significant advances in our knowledge about mechanisms that preserve genomic integrity and the identification of new molecular targets for cancer therapy. We identified one of these targets, KIF18A, in 2021 along with two other groups, and at least 10 pharmaceutical companies now have anti-KIF18A drug programs with several drugs already in clinical trials.
- Recipient of 3 nationally competitive research awards: Leukemia and Lymphoma Society Special Fellow Award, March of Dimes Basil O' Connor Starter Scholar

Research Award, and Susan G. Komen Career Catalyst Research Award, as well as multiple grants from the National Institutes of Health (NIH), including a R35-MIRA grant from NIGMS.

- Primary mentor for 1 extramural postdoctoral fellowship, 5 extramural graduate fellowships, 3 nationally competitive undergraduate research awards, and 11 locally competitive undergraduate research fellowships.
- Invited speaker at numerous international meetings, regional research meetings, and departmental seminars since 2011
- Ad hoc grant reviewer for the NIH Nuclear and Cytoplasmic Structure and Dynamics (NCSD), Biological Chemistry and Macromolecular Biophysics, and Cell and Developmental Fellowships Study Sections, as well as the French National Research Agency and the Human Frontier Science Program
- Invited reviewer for prominent journals in the fields of cell biology, developmental biology, and biophysics including: Nature, Nature Communications, EMBO Journal, Biophysical Journal, Current Biology, Developmental Cell, The Journal of Cell Biology, The Journal of Cell Science, eLife, Molecular Biology of the Cell, PNAS, and PLOS Genetics.

PROFESSIONAL SERVICE

DEPARTMENTAL SERVICE

Years	Department	Committee	Role
2012	MPBP	Graduate Program interviews	Interviewer
2012-2013	MPBP	Education Committee	Member
2013-2017	MPBP	Seminar Committee	Chair
2019-2020	MPBP	Faculty Search Committee	Member
2020-2021	MPBP	Faculty Search Committee	Member
2022-2023	MPBP	Faculty Search Committee	Chair
2023-2024	MPBP	Faculty Search Committee	Chair

COLLEGE SERVICE

Years	Service Committee	Role
2012-2016	UVM Cancer Center Grand Rounds committee	Ad hoc Reviewer
2012	UVM Cancer Center/LCCRO Scientific Review Committee	Member
2016	Strategic Faculty Committee on the Academic Structure and Funding of Basic Research	Member
2017	Faculty Research Awards ad hoc review group	Ad hoc Reviewer
2017	UVM Cancer Center Director of Major Gifts Search committee	Ad hoc Interviewer

2018-2019	New Research Building Advisory Committee	Member
2018-2024	Faculty Research Awards ad hoc review group	Ad hoc Reviewer
2018-2019	Tenure-track faculty search committee Ob/Gyn Dept	Member
2019-2020	Search Committee for Senior Associate Dean for Medical Education	Member
2025	LCOM Internal Grant Program	Ad hoc Reviewer
2025	Search Committee for Associate Dean of Undergraduate and Graduate Education & Postdoctoral Training	Chair

UNIVERSITY SERVICE

Years	Service Committee	Role
2012-2013	CMB Faculty Membership Committee	Ad hoc Member
2012-2018	CMB graduate program interviews	Interviewer
2012-2016	Financial and Physical Planning Committee	Member
2013-2016	Budget Advisory Committee to the President	Member
2013	CMB Qualifying Exam Committee	Member
2013-2017	CMB Admissions and Recruitment Committee	Member
2018-2022	CMB Admissions and Recruitment Committee	Chair
2017-2018	Tenure-track faculty search committee Dept of Biology	Member
2017-2019	UVM Postdoctoral Association	Faculty Mentor
2018-2022	CMB Steering Committee	Member
2021-2023	UVM Cancer Center Pilot Grant Review Committee	Chair
2023-present	UVM Cancer Center Ambassador	Ambassador
2023-present	UVM Cancer Center Cancer Host and Environment	Program Co-Leader
2024-present	American Cancer Society Internal Research Grant Review Panel	Co-Chair
2024-present	UVM Cancer Center Protocol Review and Monitoring Committee	Member

GOVERNMENT SERVICE

Years	Service Committee	Role
2016	NIH Nuclear and Cytoplasmic Structure and Dynamics Study Section	Ad hoc grant reviewer
2016	French National Research Agency	Ad hoc grant reviewer
2020	NIH Biological Chemistry and Macromolecular Biophysics Study Section	Ad hoc grant reviewer
2021	NIH Nuclear and Cytoplasmic Structure and Dynamics Study Section	Ad hoc grant reviewer
2022	NIH Cancer Biology special emphasis panel- ZRG1 OBT-J	Ad hoc grant reviewer
2022	NIH Cell and Developmental Biology study section- F05Q	Ad hoc grant reviewer
2024	NIH Cell and Developmental Biology study section- F05Q	Ad hoc grant reviewer

SOCIETY MEMBERSHIPS

Years	Society
1998-2004	Genetics Society of America
2007-2011	American Association for the Advancement of Science
2005-present	American Society for Cell Biology
2011-present	Full Member, University of Vermont Cancer Center

SERVICE TO PROFESSIONAL ORGANIZATIONS

Years	Service Committee	Role
2014	Human Frontier Science Program	Ad hoc grant reviewer
2016-2023	American Society for Cell Biology Finance and Audit Committee	Member
2016	Cytoskeletal Motors Gordon Research Conference	Session Chair
2017	American Society for Cell Biology Career Panel	Member
2018	Cytoskeletal Motors Gordon Research Conference	Session Chair
2021- 2023	Scientific Organization Committee for the 2023 Mitotic Spindle Croatia Meeting	Member
2022	Human Frontier Science Program	Ad hoc grant reviewer
2022	Cytoskeletal Motors Gordon Research Conference	Session Chair
2024	Novo Nordisk Foundation	Ad hoc grant reviewer
2025	Florida Breast Cancer Foundation	Ad hoc grant reviewer

SERVICE TO PROFESSIONAL PUBLICATIONS

Years	Journal/Publication/Board
2011-present	Scientific journal reviewer ¹
2015-2019	Journal of Cellular Biochemistry Editorial Board
2020-2024	Faculty Opinions Faculty Member

¹Ad hoc reviewer for the following scientific journals: Biophysical Journal, Cancer Research, Cell Reports, Current Biology, Developmental Cell, eLife, EMBO Journal, Journal of Cellular Biochemistry, Journal of Cellular Physiology, Journal of Cell Biology, Molecular and Cellular Biology, Molecular Biology of the Cell, Nature, Nature Communications, PLOS One, PLOS Genetics, PNAS, Review Commons, Science Translational Medicine

PUBLIC SERVICE

Years	Service Role	Group Served
2014	Instructor	Harwood High school students
2015	Instructor	Otter Valley High school students
2015	Guest Instructor	Shelburne Community School kindergarten and 2 nd grade students
2017	Guest Instructor	Shelburne Community School 4 th grade students
2017	Invited Speaker Women's Health and Cancer Conference	Breast cancer patients and survivors

2018	Keynote Speaker Women's Health and Cancer Conference	Breast cancer patients and survivors
2019	Instruction Supervisor for lab research workshop	Otter Valley High school students
2023	Instruction Supervisor for lab workshop	Careers in Cancer Day
2024	Instruction Supervisor for lab workshop	Careers in Cancer Day
2025	Instruction Supervisor for lab workshop	Careers in Cancer Day

SUMMARY OF SERVICE ACTIVITIES

I have provided academic service at the department, college, and university levels at UVM with an overarching goal of improving the scholarly environment within the academy. My efforts have: (1) enhanced the quality and diversity of graduate students recruited to UVM by the campus-wide Cellular, Molecular, and Biomedical Sciences program; (2) improved the research environment and infrastructure within the Larner College of Medicine (LCOM); (3) recruited new research-focused faculty to UVM within and outside of the LCOM; and (4) brought leaders in the fields of cell biology, biophysics, and cancer biology to UVM to present their research and learn about the discoveries being made by faculty here. As a co-leader of the Cancer Host and Environment Program within the UVM Cancer Center, I currently have opportunities to enhance cancer-related research efforts at the fundamental, translational, and clinical levels.

In addition to my service at UVM, I have also provided service at the national and international levels through grant and manuscript reviews, organizing and chairing sessions at international meetings, and as a member of the Finance and Audit Committee for the American Society for Cell Biology. I have also participated in or led educational outreach efforts within the local community.

TEACHING**FORMAL SCHEDULED CLASSES**

Year (semester) ¹	Course Title	Course ²		Hours ³	Number of Learners	Learner Level ⁴
		R	E			
2026 (S)	Cell and Molecular Biology (CLBI 6010)	X		5	27	G
	Undergraduate Research (MPBP 3995)	X		10	1	UG
	Undergraduate Research (MPBP 2995)		X	10	1	UG
2025 (F)	VIC- Foundations of Clinical Science	X		3	124	M
	Undergraduate Research (MPBP 2995)		X	10	1	UG
	Molecular Control of the Cell (MPBP 310/6100)		X	5	8	G
2025 (S)	Cell and Molecular Biology (CLBI 6010)	X		5	16	G
	Critical Reading and Analysis (CLBI 7010)	X		4	21	G
	Undergraduate Research (MPBP 2995)		X	30	3	UG
2024 (F)	VIC- Foundations of Clinical Science	X		3	124	M
	Undergraduate Research (MPBP 2995)		X	30	3	UG
	Molecular Control of the Cell (MPBP 310/6100)		X	5	7	G
2024 (S)	Cell and Molecular Biology (CLBI 6010)	X		5	20	G
	Critical Reading and Analysis (CLBI 7010)	X		4	15	G
2023 (F)	VIC- Foundations of Clinical Science	X		3	124	M
	Undergraduate Research (MPBP 191B)		X	20	1	UG
	Molecular Control of the Cell (MPBP 310/6100)		X	5	6	G
2023 (S)	Cell and Molecular Biology (CLBI 301)	X		8	51	G
	Cell and Molecular Biology (CLBI 301OL)	X		8	10	G
	Critical Reading and Analysis (CLBI 401)	X		4	16	G
	Undergraduate Research (MPBP 191B)		X	30	1	UG
2022 (F)	VIC- Foundations of Clinical Science	X		3	124	M
	Undergraduate Research (MPBP 191B)		X	30	1	UG
2022 (S)	Cell and Molecular Biology (CLBI 301)	X		10	58	G
	Critical Reading and Analysis (CLBI 401)	X		6	19	G
	Undergraduate Research (MPBP 191B)		X	30	3	UG
2021 (F)	VIC- Foundations of Clinical Science	X		3	124	M
	Undergraduate Research (MPBP 191B)		X	30	2	UG
	Molecular Control of the Cell (MPBP 310)		X	5	5	G
2021 (S)	Cell and Molecular Biology (CLBI 301)	X		10	43	G
	Critical Reading and Analysis (CLBI 401)	X		6	7	G
	Independent Study (MPBP 392)		X	15	1	G
	Undergraduate Research (MPBP 191B)		X	30	2	UG
2020 (F)	VIC- Foundations of Clinical Science	X		3	124	M
	Undergraduate Research (MPBP 191A)		X	30	3	UG
2020 (S)	Cell and Molecular Biology (CLBI 301)	X		10	52	G
	Critical Reading and Analysis (CLBI 401)	X		6	21	G
	Critical Reading in MPBP (MPBP 303)		X	2	5	G
	Undergraduate Research (MPBP 191B)		X	30	2	UG
2019 (F)	VIC- Foundations of Clinical Science	X		3	124	M
	Undergraduate Research (MPBP 191A)		X	30	3	UG
2019 (S)	Cell and Molecular Biology (CLBI 301)	X		3.75	40	G
	Critical Reading and Analysis (CLBI 401)	X		2	7	G
	Undergraduate Research (MPBP 191B)		X	30	3	UG
2018 (F)	VIC- Foundations of Clinical Science	X		2	120	M
	VIC- Foundations of Clinical Science- small group	X		12	10	M
	Molecular Control of the Cell (MPBP 310)		X	11	9	G

	Undergraduate Research (MPBP 191A)		X	30	3	UG
2018 (S)	Critical Reading and Analysis (CLBI 401)	X		2	8	G
	Cell and Molecular Biology (CLBI 301)	X		3.75	48	G
	Critical Reading in MPBP (MPBP 303)		X	2	2	G
	Undergraduate Research (MPBP 191B)		X	30	3	UG
2017 (F)	VIC – Foundations of Clinical Science	X		3	120	M
	VIC- Foundations of Clinical Science- small group	X		10	10	M
	Undergraduate Research (MPBP 191A)		X	30	3	UG
2017 (S)	Undergraduate Research (MPBP 191B)		X	30	3	UG
2016 (F)	VIC – Foundations of Clinical Science	X		3	120	M
	VIC- Foundations of Clinical Science- small group	X		17	10	M
	Molecular Control of the Cell (MPBP 310)		X	25	9	G
	Undergraduate Research (MPBP 191A)		X	45	4	UG
	Cell and Molecular Biology (CLBI 301)	X		3.75	48	G
2016 (S)	Undergraduate Research (MPBP 191B)		X	30	3	UG
2015 (F)	VIC – Foundations of Clinical Science	X		3	120	M
	VIC- Foundations of Clinical Science- small group	X		12	10	M
	Undergraduate Research (MPBP 191A)		X	25	2	UG
	Cell and Molecular Biology (CLBI 301)	X		3	18	G
2015 (S)	Cell and Molecular Biology (CLBI 301)	X		3	17	G
	Honors: Biochemistry (HON 276)		X	30	1	UG
	Undergraduate Research (MLSR 293)		X	15	1	UG
	Undergraduate Research (BIOC 192)		X	15	1	UG
2014 (F)	VIC – Foundations of Clinical Science	X		3	116	M
	VIC- Foundations of Clinical Science- small group	X		4	10	M
	Molecular Control of the Cell (MPBP 310)		X	16	5	G
	Honors: Biochemistry (HON 276)		X	30	1	UG
	Undergraduate Research (MLSR 293)		X	15	1	UG
	Undergraduate Research (BIOC 191)		X	15	1	UG
2014 (S)	Special Topics in Molecular Genetics	X		1	~10	G
	Honors: Biology (HON 209)		X	30	1	UG
	Advanced Undergraduate Research (BSCI 298)		X	15	1	UG
2013 (F)	VIC – Foundations of Clinical Science	X		1.5	116	M
	Cell and Molecular Biology (CLBI 301)	X		3	18	G
	Molecular Control of the Cell (MPBP 310)		X	10	11	G
	Honors: Biology (HON 208)		X	30	1	UG
2012 (F)	Cell and Molecular Biology (CLBI 301)	X		3	~20	G
	Critical Reading in MPBP (MPBP 303)		X	3	3	G
2011 (S)	Critical Reading in MPBP (MPBP 303)		X	3	3	G

¹F- fall semester; S- spring semester; ²R-required; E-elective; ³Hours in bold indicate courses for which Dr. Stumpff was Course Director. Note that only student contact hours are reported. Class preparation, grading, and director effort hours are not included in the reported time. ⁴UG-Undegraduate; G-graduate; M-Medical.

CURRICULUM DEVELOPMENT

To fill a need for an advanced cell biology class in the CMB graduate program, I designed a new course (MPBP 310: Molecular Control of the Cell) to complement the introductory cell biology course that all CMB students are required to take (CLBI 301). MPBP 310 focuses on examining fundamental molecular mechanisms that underlie a broad range of cellular processes. The material for the course comes exclusively from primary literature, and the course involves a mixture of lectures and group discussions. Instead of attempting to cover as many cellular processes as possible, this course offers an in-depth view of a few fundamental cellular processes. The emphasis is to examine molecular mechanisms that control essential biological functions at various scales ranging from single molecules to the intracellular and intercellular signaling pathways that coordinate macromolecular functions. How these mechanisms are applied during a range of other

cellular functions is then discussed. Students are evaluated in this course based on take-home exams that assess application of the material covered during class, as well as their presentations of primary literature during class. The goal of these assessments is to develop the students' critical thinking and presentation skills. MPBP 310 addresses a gap that existed in the cell biology curriculum for the CMB graduate program. Similarly, in 2016, I co-developed a primary literature-discussion based cell biology course called CLBI 401 to complement the lecture-based CLBI 301. The goal of this course is to provide first-year PhD students exposure to experimental methods and logic related to topics being covered in CLBI 301.

POSTGRADUATE AND OTHER COURSES

(see table above)

PREDOCTORAL STUDENTS SUPERVISED OR MENTORED

Years	Role	Student's Name	Program	Current Position
2025-present	Dissertation Advisor	Ethan Wagner	CMB	n/a
2023-2024	Masters Thesis Advisor	Anna Belongia	CMB	Clinical Trials Coordinator, UVM
2023-present	Dissertation Advisor	Amila Semic	CMB	n/a
2022-present	Dissertation Advisor	Sarah-Catherine Paschall	CMB	n/a
2019-2023	Dissertation Advisor	Katie Queen	CMB	Faculty, UVM Cancer Center
2017-2022	Dissertation Advisor	Alex Thompson	CMB	Clinical Variant Scientist, GeneDx
2016-2021	Dissertation Advisor	Leslie Sepaniac	CMB	Postdoc, Univ of Wisconsin
2013-2018	Dissertation Advisor	Haein Kim	CMB	Business Development Officer, Stryde Research

GRADUATE ROTATION STUDENTS MENTORED

Years	Role	Student Name	Program
2026-2027	Laboratory Rotation Advisor	Porter Tomsick	CMB
2026	Laboratory Rotation Advisor	Abhi Ananthasuryanarayana	CMB
2024-2025	Laboratory Rotation Advisor	Ethan Wagner	CMB
2024	Laboratory Rotation Advisor	Charlie Watts	CMB
2022-2023	Laboratory Rotation Advisor	Amila Semic	CMB
2022	Laboratory Rotation Advisor	Anna Belongia	CMB
2022	Laboratory Rotation Advisor	Allison Racela	CMB
2021-2022	Laboratory Rotation Advisor	Sarah-Catherine Paschall	CMB

2020	Laboratory Rotation Advisor	Randi Gravelle	CMB
2018-2019	Laboratory Rotation Advisor	Katie Queen	CMB
2017	Laboratory Rotation Advisor	Scott Vanson	CMB
2016-2017	Laboratory Rotation Advisor	Alex Thompson	CMB
2016	Laboratory Rotation Advisor	Kathryn Svec	CMB
2015-2016	Laboratory Rotation Advisor	Christopher Villa	CMB
2015	Laboratory Rotation Advisor	Leslie Sepaniac	CMB
2013	Laboratory Rotation Advisor	Joseph Clayton	CMB
2012-2013	Laboratory Rotation Advisor	Haein Kim	CMB
2012	Laboratory Rotation Advisor	Joyce Thompson	CMB

MEDICAL STUDENTS SUPERVISED OR MENTORED

Years	Role	Student Name	Program
2026-present	Research Mentor	Stephanie Miller	UVM Medical Student
2021-2022	Mentor in LCOM Mentor Program for students from under-represented groups	Dario Vanegas Betancur	UVM Medical Student

UNDERGRADUATE STUDENTS SUPERVISED OR MENTORED

Years	Role	Student Name	Major	Current Position
2026	STEM Scholar/ Research Advisor	Naomi Lyons		UVM undergrad
2026	STEM Scholar/ Research Advisor	Mollie Delacruz		UVM undergrad
2026	STEM Scholar/ Research Advisor	Lance Francisco		UVM undergrad
2026	Summer Undergraduate Research Advisor	Peter Gilliam*	Chemistry	Hobart College undergrad
2026	Summer Undergraduate Research Advisor	Ashley Lefebvre*	Biochemistry	St. Michael's College undergrad
2025-present	STEM Scholar/ Research Advisor	Elsa Williams	Neuroscience	UVM undergrad
2025-2026	STEM Scholar/ Research Advisor	Orla Murphy	MMG	UVM undergrad

2025	STEM Scholar Advisor	Kai Dirksen	Biochemistry	UVM undergrad
2024-2025	Honors Thesis Advisor	Ava Caito	MMG	EMT, Boston
2024-2025	Summer Undergraduate Research Advisor	Turner Elliott*	Chemistry (Haverford College)	Haverford College Undergrad
2024-2025	Summer Undergraduate Research Advisor	Lauren Tuzic*	Biology (Norwich University)	Dartmouth Grad Student
2023-present	Masters Research Advisor	Lily Kearns	MMG	UVM undergrad
2023-2024	Summer Undergraduate Research Advisor	Maxine Robinson*	Engineering Physics (Delaware St.)	PhD student, Physics, Delaware St.
2023-2025	Undergraduate Research Advisor	Petra Kapsalis*	MMG	Medical Assistant, Colorado
2022-2023	Laboratory Work Study Mentor	Ryder McLaughlin	Biology	UVM undergrad
2021-2023	Honors Thesis Advisor	Olivia Budington*	Biology	Medical Assistant, VT
2021-2022	Laboratory Work Study Mentor	Kira Fisher	MMG	Senior Research Technician, UVM (Krag Lab)
2021-2022	Undergraduate Research Advisor	Noah Arons	Animal Science	Vet Student, Univ of Glasgow
2019-2022	Honors Thesis Advisor	Hannah Poquette*	Biochemistry	Research Lab Technician, Green Mountain Antibodies
2019- 2020	Laboratory Work Study Mentor	Jade Rigout	Bioengineering	Senior Consultant, Praecipio Consulting
2019	Undergraduate Summer Research Advisor	Samantha Provost	Biology (Wake Forest)	PhD student at Univ of Pennsylvania
2018-2021	Undergraduate Research Advisor	Sarah Stevens	Undecided	Certified Nursing Assistant
2018- 2019	Honors Thesis Advisor	Kyra Fryling	MMG	Senior Associate, Pew Charitable Trusts
2017-2018	Laboratory Work Study Mentor	Thea Pappas	Neuroscience	Patient Health and Wellness Coach, CommuniCare
2017-2018	Undergraduate Research Advisor	Sarah Vandal*	Neurosc (Colby)	PhD student, Dartmouth College

2017	Undergraduate Summer Research Advisor	Maria Kogan	Computer Science (Columbia)	Technical Staff III at VMware (Palo Alto, CA)
2016-2019	Honors Thesis Advisor	Carolyn Marquis*	MMG	Medical Student, Brown University
2016-2019	Undergraduate Research Advisor	Christopher Kruglik	Biochemistry	Medical student, UVM
2016-2018	Honors Thesis Advisor	Lisa Wood*	MMG	PhD student, Cancer Biology, CU Denver
2016-2017	Undergraduate Research Advisor	Jillian Kelly	Med Lab Science	Medical Lab Scientist, UVM
2014-2017	Honors Thesis Advisor	Dana Messinger*	Med Lab Science	PhD student, Cancer Biology, Univ of Michigan
2014-2015	Honors Thesis Advisor	Julia Torvi*	Biochemistry	PhD from Berkeley, Scientist at The Allen Institute (Seattle, WA)
2014	Undergraduate Research Advisor	Cooper Pearson	Biology	Senior Software Engineer at Optum
2013-2014	Honors Thesis Advisor	Samantha Bissonette	Biology	Emergency Medicine Resident, Dartmouth
2012-2016	Laboratory Work Study Mentor	Sean Lenahan	Biochemistry	Postdoc, Harvard

*Students awarded competitive summer research fellowships under Dr. Stumpff's mentorship.

CURRENT DISSERTATION/THESIS COMMITTEE MEMBERSHIP at UVM

Years	Role	Student Name	Program
2026-present	Dissertation Committee Member	Selina Mintaah	CMB
2026-present	Dissertation Committee Member	Alexa Price	CMB
2025-present	Dissertation Committee Member	Madeline McTigue	CMB
2024-present	Dissertation Committee Member	Allison Racela	CMB
2024-present	Dissertation Committee Chair	Vanessa Lopez	CMB
2023-present	Dissertation Committee Member	Daniel Richter	Biology
2022-present	Dissertation Committee Member	Allison Morrissey	CMB

COMPLETED GRADUATE DISSERTATION/THESIS COMMITTEES at UVM

Years	Role	Student Name	Program
2023-2024	Masters Thesis Committee Member	Priyanka Chavarkar	Pharmacology
2022-2024	Dissertation Committee Member	Erin Gaston	CMB
2022-2024	Masters Thesis Committee Chair	Phoebe Cousens	Biology
2019-2024	Dissertation Committee Member	Frances Male	CMB
2017-2024	Dissertaton Committee Member	Mingu Kang	CMB
2017-2022	Dissertaton Committee Member	Inessa Manuelyan	CMB
2018-2022	Dissertation Committee Member	Rachel Stadler	CMB
2018-2022	Dissertaton Committee Member	Kathryn Svec	CMB
2018-2022	Dissertaton Committee Member	Alisa Cario	CMB
2018-2022	Dissertaton Committee Member	Scott Vanson	CMB
2017-2022	Dissertaton Committee Member	Sierra Bruno	CMB
2017-2020	Dissertaton Committee Member	Dominique Lessard	CMB
2016-2019	Dissertation Committee Chair	Riley St. Clair	NGP
2015-2019	Dissertation Committee Member	Blas Guigni	CMB
2017-2018	Dissertaton Committee Member	Hannah Naughton	CMB
2013-2018	Dissertation Committee Member	Jamie Stern	CMB
2015-2018	Dissertation Committee Member	Rehan Ali	NGP
2013-2017	Dissertation Committee Member	Benjamin King	CMB
2016	Dissertation Committee Chair	Alden Clements	Chemistry
2015-2017	Dissertation Committee Member	Miranda Redmond	CMB
2015	Dissertation Committee Chair	Shruthi Krishnamurthy	MMG
2014-2016	Dissertation Committee Chair	Ryan Joy	Biology
2013-2016	Dissertation Committee Member	Joe Clayton	CMB
2013-2014	Dissertation Committee Member	Laura Director	CMB
2012-2017	Dissertation Committee Member	Luther Pollard	CMB
2012-2016	Dissertation Committee Member	Greg Hoeprich	MPBP
2012-2014	Dissertation Committee Member	Andrew McKenzie	Pharmacology

GRADUATE STUDENT DISSERTATION/THESIS COMMITTEES OUTSIDE UVM

Years	Role	Student Name	Institution
2023-2024	Dissertation Committee Member	Byron Hunter	PhD at Queens University
2023-2024	Dissertation Committee Member	Leela Biswas	MD/PhD at Rutgers Univ

2020	Dissertation Committee Member	Ryan Quinton	MD/PhD at Boston Univ
2019	Dissertation Committee Member	Conrad Hall	PhD at McGill Univ
2018	Dissertation Committee Member	Kelly Salmon	PhD at Dartmouth
2014	Dissertation Committee Member	Marianna Kleyman	PhD at Dartmouth
2013	Thesis Committee Member	David Bing	MS at Laurentian Univ

POSTDOCTORAL FELLOWS AND RESIDENTS DIRECTLY SUPERVISED OR MENTORED

Years	Role	Fellow's Name	Ph.D./M.D Institution	Current Position
2025-present	Fellowship Research Advisor	Lindsey Alpeter, M.D.	Wake Forest	REI Fellow, UVM
2024-2026	Fellowship Research Advisor	Navin Maredia Ebrahimi, M.D.	UT Southwestern	Ob/Gyn, Chicago, IL
2022-2024	Fellowship Research Advisor	Carleigh Nesbit, M.D.	University of New England	Physician at Boston IVF
2020-2022	Postdoctoral Advisor	Katherine Schutt, Ph.D.	Dartmouth College	Cell Biologist at Topos Bio
2018-2020	Fellowship Research Advisor	Jessica Ryniec, M.D.	Georgetown	Physician at CCRM Boston Fertility
2014-2019	Postdoctoral Advisor	Heidi Malaby, Ph.D.	UMass Med	Team Lead for Complex Proposals and Strategic Positioning, UVM

FACULTY MENTORED

Dates	Name	Position while Mentored	Faculty Role	Current Position
2025-present	Matthew Hannaford	Assistant Professor	Faculty Mentor	Assistant Professor
2021-present	John Salogiannis, PhD	Assistant Professor	Faculty Mentor	Assistant Professor
2020-2023	Nimrat Chatterjee, Ph.D.	Assistant Professor	Faculty Mentor	Assistant Professor
2019-2024	Masayo Koide, Ph.D.	Assistant Professor	Faculty Mentor	Assistant Professor
2016-2019	Michael Previs, Ph.D.	Assistant Professor	Faculty Mentor	Associate Professor

HONORS/AWARDS TO POSTDOCTORAL MENTEES:

Year(s)	Postdoc Name	Award
2017-2019	Heidi Malaby	Horizon Award from the Department of Defense Peer Reviewed Cancer Research Program (~\$60K/ year)

HONORS/AWARDS TO GRADUATE MENTEES:

Year(s)	Student Name	Award
2013	Haein Kim	CMB Best Poster Presentation Award
2014	Haein Kim	Larner College of Medicine Graduate Student Symposium Best Poster Presentation Award
2017	Leslie Sepaniac	Larner College of Medicine Dean's Excellence in Research Award- 1 st Place talk in Junior Division
2017	Alex Thompson	Larner College of Medicine Dean's Excellence in Research Award- 2 nd Place talk in Junior Division
2017	Alex Thompson	MPBP Cathy Bulman Memorial Award for Best Poster
2018	Alex Thompson	Larner College of Medicine Dean's Excellence in Research Award- 1 st Place talk in Junior Division
2018	Leslie Sepaniac	Larner College of Medicine Dean's Excellence in Research Award- 2 nd Place talk in Senior Division
2018	Leslie Sepaniac	CMB Director's Award in recognition of service
2020	Alex Thompson	Larner College of Medicine Dean's Excellence in Research Award- 1 st Place talk in Senior Division
2020	Leslie Sepaniac	Larner College of Medicine Dean's Excellence in Research Award- 2 nd Place talk in Senior Division
2020	Alex Thompson	MPBP Cathy Bulman Memorial Award for Most Inquisitive Questions
2017-2021	Leslie Sepaniac	Vermont Space Grant Consortium Graduate Fellowship (\$35K/year)
2019-2022	Alex Thompson	National Institutes of Arthritis and Musculoskeletal and Skin Diseases (NIAMS) F31 Individual Fellowship (\$32K/year)
2020-2023	Katie Queen	National Science Foundation Graduate Fellowship (\$40K/year)
2021	Leslie Sepaniac	Larner College of Medicine Most Outstanding Publication from a Graduate Student
2022	Alex Thompson	Larner College of Medicine Most Outstanding Publication from a Graduate Student
2024	Amila Semic	MPBP Cathy Bulman Memorial Award for Best Poster
2024	Sarah-Catherine Paschall	Vermont Space Grant Consortium Graduate Fellowship (\$35K/year)
2024	Amila Semic	Graduate Assistance in Areas of National Need (GAANN) Fellowship (\$35K/year)

HONORS/AWARDS TO UNDERGRADUATE MENTEES:

Year(s)	Student Name	Award
2014	Samantha Bisonnette	UVM Honors Thesis in Biology
2014	Julia Torvi	Woodworth Award Summer Fellowship from UVM Biochemistry
2015	Dana Messinger	UVM Summer Undergraduate Research Fellowship (SURF)
2016	Dana Messinger	UVM Honors Thesis in Medical Laboratory Sciences
2016	Lisa Wood	UVM Summer Undergraduate Research Fellowship (SURF)
2017	Carolyn Marquis	UVM Summer Undergraduate Research Fellowship (SURF)
2017	Sarah Vandal	National Science Foundation Summer Neuroscience Undergraduate Research Fellowship
2017	Lisa Wood	Distinguished Undergraduate Research Award Microbiology and Molecular Genetics
2018	Lisa Wood	Lucille P. Markey Outstanding Senior in Molecular Genetics Award (Thesis)
2018	Carolyn Marquis	National Science Foundation Summer Neuroscience Undergraduate Research Fellowship
2019	Kyra Fryling	Distinguished Undergraduate Research Award Microbiology and Molecular Genetics (Honors College Thesis)
2019	Carolyn Marquis	Distinguished Undergraduate Research Award Microbiology and Molecular Genetics (Honors College Thesis)
2019	Carolyn Marquis	Lucille P. Markey Outstanding Senior in Molecular Genetics Award
2020	Hannah Poquette	UVM Biochemistry Summer Undergraduate Research Fellowship (Thesis)
2021	Hannah Poquette	UVM Summer Undergraduate Research Fellowship (SURF)
2022	Hannah Poquette	Distinguished Undergraduate Research Award in Biochemistry
2023	Olivia Budington	Honors College Thesis
2023	Olivia Budington	UVM Joan M. Herbers Award in Biology for outstanding research accomplishments
2023	Petra Kapsalis	UVM Cancer Center Summer Research Fellowship
2023	Maxine Robinson	Vermont Biomedical Research Network Summer Research Fellowship
2024	Petra Kapsalis	Nicole J Ferland Award from for Summer Undergraduate Research
2024	Lauren Tuzik	Vermont Biomedical Research Network Summer Research Fellowship
2025	Ava Caito	Distinguished Undergraduate Research Award in Biology
2025	Petra Kapsalis	Distinguished Undergraduate Research Award in MMG
2025	Petra Kapsalis	Warren R. Stinebring Outstanding Senior in Microbiology Award
2025	Turner Elliott	Koshland Integrated Natural Sciences Center (KINSC) Summer Fellowship

SUMMARY OF TEACHING ACTIVITIES

As a member of the Molecular Physiology and Biophysics department, I am expected to contribute to the graduate and medical education missions of the College of Medicine. I have made significant contributions to graduate teaching in the classroom, in training laboratory scientists, and as a member of dissertation committees. I have a strong background in cell biology and I am involved in the annual review and revision of core graduate instruction in cell biology as the director of MPBP 6100 and a regular instructor in CLBI 6010. A primary challenge for our initial graduate cell biology course is how to make the material relevant and engaging for the mix of PhD and Masters students who are required to enroll in this class. A couple of the most significant changes made to address this problem in recent years were to incorporate active learning approaches with the use of clicker questions and peer-to-peer instruction and to develop a complementary primary literature course (CLBI 7010) to provide CMB PhD students with opportunities to explore experimental design and discovery in greater detail.

In fall of 2013, I began teaching first-year medical students through the Vermont Integrated Curriculum Cell and Molecular Biology course, which has since changed its name to Foundations of Medicine and then Foundations of Clinical Science. Over the last 11 years, I have worked to improve my teaching skills in this course, which typically has 124 students. To that end, I have invited senior faculty to attend my lectures and provide critical feedback. By incorporating the suggestions of these mentors, my ability to communicate the material to a diverse set of students has improved, and I consistently receive high course evaluation scores.

While it is not required, I have additionally been involved in undergraduate teaching via mentoring students conducting independent research projects in my lab. I view active laboratory research as a key strength that can be leveraged to recruit top flight undergraduate students to the University of Vermont. Thus, I have actively worked to provide research opportunities to undergraduates within my research program and offer an undergraduate course in independent research (MPBP 191). Since the fall of 2013, I have had one to four undergraduate students working in my lab for credit each semester (total of 27 students). 8 of these students have successfully completed honors thesis projects, and 10 have won competitive fellowships to support full time research in my lab during the summer. While one-on-one teaching in the lab requires a considerable level of effort per student, I believe these opportunities significantly enhance the educational experience of science majors.

RESEARCH AND SCHOLARLY ACTIVITIES

RESEARCH AWARDS AND GRANTS

Ongoing Research Support

R35GM144133 (Stumpff, PI)

02/01/22-01/31/27

NIH/ NIGMS

\$276,000 direct/ year

“Mechanisms of microtubule motors and chromosome segregation”

The primary goal of this project is to determine the molecular mechanisms utilized by kinesin motor proteins to control mitotic spindle function.

Role: PI (45%)

Juckett Faculty Scholar Award

07/01/26-06/30/28

J. Walter Juckett Foundation

\$150,000 direct/ year

This award does not have specific project objectives. Juckett Faculty Scholars will receive \$150,000 per year for two years to support their research activities. No less than half of this amount is to be utilized for Juckett Faculty Scholar salary/fringe benefit support. The remainder is unrestricted and can be utilized for research staff, student/trainee support, research supplies, shared resources services and other expenses.

Role: PI (25%)

F31 AR087752-01 (Semic, PI)

06/01/26-05/31/28

NIH/ NIAMS

\$35,202 direct/ year

“Investigating the contributions of chondrocyte spindle biophysics and tissue microenvironment to skeletal dysplasia”

This competitive, NIH funded fellowship supports the stipend and tuition of Amila Semic, a Ph.D. student in Dr. Stumpff’s laboratory. The primary goal of the project is to determine how mutations in KIF22 lead to bone dysplasia.

Role: Mentor

Completed Research Support

Administrative Supplement for R35GM144133

02/01/24-01/31/25

NIGMS

\$250,000 direct total

This award provided funds to purchase a spinning disk confocal microscope.

Scientific Research Agreement (Stumpff, PI)

05/15/22-05/14/24

Aperion Therapeutics

\$388,560 direct total

“Investigating potential anti-cancer drugs”

The primary goal of this project is to investigate small molecule inhibitors for their ability to limit tumor cell proliferation.

Role: PI (10%)

R01GM130556 (Rosenfeld, Stumpff, Muretta, Gross, MPIs)

03/15/19-01/31/23

NIH/NIGMS

\$91,000 direct/year

Currently in NCE

“Tuning mitotic kinesins through motor domain post-translational modifications”

The primary goal of this project is to understand how the post-translational modification of mitotic kinesins affects their activity and function from the single molecule to whole cell scale.

Role: MPI (10% effort)

Administrative Supplement for R35GM144133

07/01/22-06/30/23

NIGMS

\$250,000 direct total

This award provided funds to purchase a ring TIRF microscope.

Scientific Research Agreement (Stumpff, PI)

07/01/22-04/30/23

Volastra Therapeutics

\$30,000 direct total

“Single molecule investigation of anti-kinesin drugs”

The primary goal of this project is to investigate the effects of small molecule inhibitors on kinesin function in single molecule assays.

Role: PI (5%)

R01GM121491 (Stumpff, PI)

09/05/17-06/30/22

NIH/NIGMS

\$200,000 direct/year

“Spatial and temporal control of mitotic chromosome movements”

The goal of this project is to determine the mechanisms underlying mechanical control of chromosome alignment in normal cells.

Role: PI (30% effort)

Scientific Research Agreement (Stumpff, PI)

07/01/21-06/30/22

Aperion Therapeutics

\$27,120 direct total

“Investigating potential anti-cancer drugs”

The primary goal of this project is to test small molecule inhibitors for their ability to limit tumor cell proliferation.

Role: PI (2% effort)

UVMCC Pilot Award (Stumpff, MPI)

07/01/20-06/30/21

University of Vermont Cancer Center

\$50,000 direct/ year

“Investigating the impact of micronuclei on genomic stability of primary tumor cells ”

The primary goal of this project is to determine whether micronuclei alter genomic stability in vivo.

Role: MPI (5% effort)

Administrative Supplement for R01GM121491 (Stumpff, PI)

07/01/19-06/30/20

NIH/NIGMS

\$200,000 direct/year

This award provided funds to purchase an advanced microscopy system for live cell imaging.

Administrative Supplement for R01GM121491 (Stumpff, PI)

07/01/18-06/30/19

NIH/NIGMS

\$50,805 direct/year

This award provided funds to purchase an automated microscopy system.

CCR16377648 Career Catalyst Research Grant (Stumpff, PI)

09/14/16-09/13/20

Susan G Komen

\$120,000 direct/ year

“A Novel Strategy for Anti-mitotic Therapy in Breast Cancer”

The primary goal of this project is to determine the suitability of nonessential mitotic proteins as therapeutics for triple negative breast cancers.

Role: PI (5% effort)

VCC/ LCCRO Pilot Award (Stumpff, Howe, Co-PIs)

05/01/17-04/30/18

Vermont Cancer Center/ LCCRO

\$65,000 direct

“A mechanistic study of kinesin function during triple negative breast cancer cell migration”

The primary goal of this pilot project was to determine the mechanistic contribution of KIF18A to the migration of triple negative breast cancer cells.

Role: PI (10% effort)

VCC/ LCCRO Pilot Award (Stumpff, Howe, Anker, Co-PIs)

07/01/15-06/30/16

Vermont Cancer Center/ LCCRO

\$50,000 direct

“Investigating Kif18A as a Therapeutic Target for Cancer”

The primary goal of this project was to determine the suitability of Kif18A and non-essential mitotic proteins as anti-cancer targets and radiosensitizers for colon and breast cancer.

Role: PI (10% effort)

ACS Institutional Research Grant (Stumpff, Anker Co-PIs)

06/01/15-05/31/16

The American Cancer Society \$30,000 direct

“Investigating Kif18A as a Therapeutic Target for Colorectal Cancer”

The primary goal of this project was to determine the suitability of Kif18A and non-essential mitotic proteins as anti-cancer targets and radiosensitizers.

Role: PI (10% effort)

Basil O'Connor Starter Scholar Research Award (Stumpff, PI) 02/01/14-01/31/16

The March of Dimes Foundation \$75,000 total/ year

“Mechanical Control of Chromosome Geometry”

The primary goal of this project was to determine the mechanical mechanisms that control the orientation of paired chromosomes during cell division and promote their equal distribution.

Role: PI (30% effort)

VCC/ LCCRO Program Award (Stumpff, Tang Co-PIs) 12/15/13-12/14/14

Vermont Cancer Center/ LCCRO \$25,000 direct

“Mapping the Effects of Mitotic Chromosome Organization on Interphase Genomic Architecture”

The primary goal of this one year pilot award was to investigate the effects of mitotic chromosome alignment defects, caused by loss of Kif18A function, on chromosome segregation accuracy and the 3D organization of the interphase genome.

Role: PI (10% effort)

Special Fellow Award (Stumpff, PI) 07/01/10-06/30/13

Leukemia and Lymphoma Society \$65,000 direct/ year

“Regulation of Mitosis by the Shwachman-Bodian-Diamond Syndrome Protein”

The primary goal of this project was to investigate the functions of SBDS in promoting successful mitotic cell division.

Role: PI (45% effort)

Individual NRSA Postdoctoral Fellowship (Stumpff, PI) 09/16/06-09/15/08

NIH/NIGMS \$60,000 direct/ year

“Kinesin-regulated microtubule dynamics”

This project involved investigation of Kif18A's function during mitosis and led to the discovery that it is a key regulator of chromosome movements and alignment.

Role: PI (100% effort)

Primary Mentor for Completed External Fellowships

VTSGC Graduate Fellowship (Paschall, PI) 07/01/24-06/30/25

Vermont Space Grant Consortium ~\$35,000 direct/ year

“Molecular Mechanisms of KIFBP in promoting neuronal development and maintenance”

This competitive, NASA funded fellowship supports the stipend and tuition of Sarah-Catherine Paschall, a Ph.D. student in Dr. Stumpff's laboratory. The primary goal of the project is to determine how KIFBP regulates the cytoskeleton during neuronal differentiation.

Role: Mentor

CMB-GAANN Fellowship (Semic, PI) 07/01/24-06/01/26

US Department of Education ~\$40,000 direct/ year

“Investigating the Contributions of Chondrocyte Spindle Biophysics and Tissue Microenvironment to Skeletal Dysplasia”

This is a competitive fellowship from the Graduate Assistance in Areas of National Need program of the US Department of Education. The fellowship supports the stipend and tuition of Amila Semic, a Ph.D. student in Dr. Stumpff's laboratory. The primary goal of the project is to determine how mutations in KIF22 cause skeletal dysplasia.

Role: Mentor

NSF Graduate Research Fellowship (Queen, PI)

09/01/20-08/31/23

National Science Foundation

~\$40,000 direct/year

"Regulation of kinesin by post-translational modifications during mitosis"

This competitive fellowship supported the stipend and tuition of Katie Queen, a Ph.D. student in Dr. Stumpff's laboratory. The primary goal of the project was to determine how post-translational modifications within the enzymatic region of KIF18A regulate its mitotic functions.

Role: Mentor

F31 AR074887 (Thompson, PI)

07/01/19-06/30/22

NIH/ NIAMS

\$32,745 direct/ year

"Molecular etiology of Spondyloepimetaphyseal dysplasia with joint laxity, leptodactylic type"

This competitive, NIH funded fellowship supported the stipend and tuition of Alex Thompson, a Ph.D. student in Dr. Stumpff's laboratory. The primary goal of the project was to determine how mutations in KIF22 cause bone development defects.

Role: Mentor

VTSGC Graduate Fellowship (Sepaniac, PI)

07/01/17-05/31/21

Vermont Space Grant Consortium

~\$35,000 direct/ year

"Mechanisms of micronuclear envelope rupture"

This competitive, NASA funded fellowship supported the stipend and tuition of Leslie Sepaniac, a Ph.D. student in Dr. Stumpff's laboratory. The primary goal of the project was to determine how micronuclei impact the genomic stability of human cells.

Role: Mentor

CA160988 Horizon Award (Malaby, PI)

09/01/17-08/30/19

Department of Defense, Peer Reviewed Cancer Research Program

~\$60,000 direct/ year

"Mechanisms of selective susceptibility to inhibition of a cytoskeletal regulator in colorectal cancer cells"

This nationally competitive fellowship supported the research efforts and career development of Dr. Heidi Malaby, a postdoctoral fellow in Dr. Stumpff's laboratory. The project aimed to understand the dependence of colorectal cancer cell division and migration on the kinesin-like motor protein KIF18A.

Role: Mentor

SCHOLARSHIP

Peer Reviewed Publications- *H-Factor=24; Total Citations >2250*

Original Research

1. Su TT, Walker J, and Stumpff J (2000). Activating the DNA damage checkpoint in a developmental context. *Curr Biol* 10:119-126. PMID: 10679321.

2. Stumpff J, Duncan T, Homola E, Campbell SD, and Su TT (2004). *Drosophila* Wee1 kinase regulates Cdk1 and mitotic entry during embryogenesis. *Curr Biol* 14:2143-2148, PMID: PMC3242732. (F1000)
3. Stumpff J, Kellogg D, Krohne K, and Su TT (2005). *Drosophila* Wee1 interacts with the γ TURC and is required for proper mitotic spindle morphogenesis and positioning. *Curr Biol* 15:1525-1534, PMID: PMC3242738.
4. Stumpff J, von Dassow G, Wagenbach M, Asbury C and Wordeman L (2008). The kinesin-8 motor Kif18A suppresses kinetochore movements to control mitotic chromosome alignment. *Dev Cell* 14:252-262, PMID: PMC2267861. (F1000)
5. Garcia K, Stumpff J, Duncan T and Su TT (2009) Tyrosines in the Kinesin-5 Head Domain are necessary for phosphorylation by Wee1 and for mitotic spindle integrity. *Curr Biol* 19:1-7, PMID: PMC2762001.
6. Mattison CP, Stumpff J, Wordeman L and Winey M (2011). Mip1 associates with both the Mps1 kinase and actin, and is required for cell cortex stability and anaphase spindle positioning. *Cell Cycle* 10:783-793, PMID: PMC3100791. (featured on journal **cover**)
7. Stumpff J*, Du Y*, English CA, Wagenbach M, Hyman AA, Asbury CL, Wordeman L and Ohi R (2011). A tethering mechanism controls the processivity and kinetochore-microtubule plus-end enrichment of the kinesin-8 Kif18A. *Mol Cell* 43:764-775, PMID: PMC3172727. (*equal contribution) (F1000)
8. Stumpff J*, Wagenbach M, Franck A, Asbury CL and Wordeman L* (2012). Kif18A and chromokinesins confine centromere movements via microtubule growth suppression and spatial control of kinetochore tension. *Dev Cell* 22: 1017-1029, PMID: PMC3356572. (*co-corresponding authors)
9. Cunniff B, Benson K, Stumpff J, Newick K, Held P, Taatjes D, Joseph J, Kalyanaraman B, Heintz NH (2013) Mitochondrial-targeted nitroxides disrupt mitochondrial architecture and inhibit expression of peroxiredoxin 3 and FOXM1 in malignant mesothelioma cells. *J Cell Physiol* 228:835-845. PMID: PMC3928986.
10. Cunniff B, Snider GW, Fredette N, Stumpff J, Hondal RJ and Heintz NH (2014). Resolution of oxidative stress by thioredoxin reductase: Cysteine versus selenocysteine. *Redox Biol.* 2:475-484. PMID: PMC3949094.
11. Kim H, Fonseca C and Stumpff J (2014). A unique kinesin-8 surface loop provides specificity for chromosome alignment. *Mol Biol Cell* 21: 3319-3329. PMID: 25208566.
12. Czechanski A, Kim H, Byers C, Greenstein I, Stumpff J*, and Reinholdt LG* (2015). Kif18a is specifically required for mitotic progression during germ development. *Dev Biol.* 402: 253-262. PMID: PMC4450139 (*co-corresponding authors)
13. Zhou J, Chan J, Lambele M, Yusufzai T, Stumpff J, Opresko PL, Thali M, and Wallace SS (2017). NEIL3 repairs telomere damage during S phase to secure chromosome segregation at mitosis. *Cell Reports.* 20:2044-2056. PMID: 28854357.
14. Muretta JM*, Reddy BJN*, Scarbelli G*, Thompson AF*, Jariwala S, Major J, Venere M, Rich JN, Willard B, Thomas DD, Stumpff J, Grant BJ, Gross SP, and Rosenfeld SS (2018). A post-translational modification of Eg5 that enhances its mechano-chemical coupling and alters its mitotic function. *PNAS.* 115:E1779-E1788 (*equal contribution) PMID: PMC5828613

15. Tracy K, Tye C, Ghule P, Malaby H, Stumpff J, Stein J, Stein G, and Lian J (2018). Mitotically associated long non-coding RNA MANCR affects genomic stability and cell division in aggressive breast cancer. *Mol Cancer Res.* 16(4): 587-598. PMID: PMC5882506
16. Malaby HLH, Lessard DV, Berger CL, and Stumpff J (2019). KIF18A's neck linker permits navigation of microtubule-bound obstacles within the mitotic spindle. *LSA.* 2019, 2:e201800169. PMID: PMC6337737
17. Malaby HLH, Dumas M, Ohi R, and Stumpff J (2019). Kinesin binding protein ensures accurate chromosome segregation by buffering Kif18A and Kif15. *JCB* 218: 1218-1234, PMID: PMC6446846 (*co-corresponding authors).
18. Fonseca CL, Malaby HLH, Sepaniac LA, Martin W, Byers C, Czechanski A, Messinger D, Tang ME, Ohi R, Reinholdt LG, and Stumpff J (2019). Mitotic chromosome alignment ensures mitotic fidelity by promoting interchromosomal compaction during anaphase. *JCB* 218: 1148-1163, PMID: PMC6446859 (*co-corresponding authors)
 - This work was highlighted in Orr B and Maiato H. (2019). No chromosome left behind: The importance of metaphase alignment for mitotic fidelity. *JCB* 218: 1086-1088, PMID: PMC6446845.
19. Bodrug T, Wilson-Kubalek EM, Nithianantham S, Thompson AF, Alfieri A, Gaska I, Major J, Debs G, Inagaki S, Gutierrez P, Gheber L, McKenney RJ, Sindelar CV, Milligan R, Stumpff J, Rosenfeld SS, Forth ST, and Al-Bassam J (2020). The kinesin-5 tail domain directly modulates the mechanochemical cycle of the motor domain for anti-parallel microtubule sliding. *eLife* 9:e511131. PMID: PMC7015671.
20. Cohen-Sharir Y, McFarland JM, Abdusamad M, Marquis C, Tang H, Ippolito MR, Bernhard SV, Laue K, Malaby HLH, Jones A, Kazachkova M, Lyons N, Nagaraja A, Bass AJ, Beroukhim R, Santaguida S, Stumpff J, Golub TR, Storchova Z, and Ben-David U. (2021). Aneuploidy renders cancer cells vulnerable to mitotic checkpoint inhibition. *Nature* 590: 486-491. PMID: PMC8262644
21. Marquis C, Fonseca CL, Queen KA, Wood L, Vandal SE, Malaby HLH, Clayton JE, and Stumpff J. (2021). Chromosomally unstable tumor cells specifically require KIF18A for proliferation. *Nat Comm.* 12: 1213. PMID: PMC7900194
22. Sepaniac LA, Martin W, Dionne LA, Stearns TM, Reinholdt LG*, and Stumpff J* (2021) Micronuclei in Kif18a mutant mice form stable micronuclear envelopes and do not promote tumorigenesis. *JCB* 220 (11). PMID: PMC8441830 (*co-corresponding authors)
23. Solon AL*, Tan Z*, Schutt KL, Jepsen L, Haynes SE, Nesivizhskii AI, Sept D, Stumpff J, Ohi R, and Cianfrocco MA. (2021) Kinesin-binding protein remodels the kinesin motor to prevent microtubule-binding. *Science Advances* 7(47). PMID: PMC8604404 (*equal contribution).
24. Thompson AF, Blackburn PR, Babovic-Vuksanovic D, Lian JB, Klee EW, and Stumpff JK. (2022) Pathogenic mutations in the chromokinesin KIF22 disrupt anaphase chromosome segregation. *Elife* 11 (e78653). PMID: PMC9302971.
25. Queen, KA, Cario, A, Berger CL, and Stumpff J. (2024) Modification of the neck linker of KIF18A alters microtubule subpopulation preference. *MBoC* 35(1). PMID: PMC10881168

26. Schutt K, Queen, KA, Fisher K, Budington O, Wei M, Xiao Y, Aswad F, Joseph J, and Stumpff J. (2024) Identification of the KIF18A alpha-4 helix as a therapeutic target for chromosomally unstable tumor cells. *Front. Mol. Biosci.* 11:1328077. PMID: PMC10896213
27. Day CA, Grigore F, Hakkim FL, Paul S, Langfald A, Weberg M, Fadness S, Schwab P, Sepaniac L, Stumpff J, Vaughan KT, Daniels DJ, Robinson JP, and Hinchcliffe EH. (2025) The histone H3.3 K27M mutation suppresses Ser31 phosphorylation and mitotic fidelity, which can drive gliomagenesis. *Current Biology.* 35:354-362. PMID: 39729988
28. Phillips AF, Zhang R, Jaffe M, Schulz R, Verma A, Feinberg T, Arensman M, Chiu A, Letso R, Bosco N, Carty MC, Bettigole S, Andreu C, Drutman S, Queen KA, Racela AR, Stumpff J, Su M, Cogan D, Eng CH. (2025) Targeting chromosomally unstable tumors with a selective KIF18A inhibitor. *Nature Communications.* PMID: 39747049
29. Nesbit C, Martin W, Czechanski A, Byers C, Raghupathy N, Ferraj A, Stumpff J, and Reinholdt L. (2025). Anapc5 and Anapc7 as genetic modifiers of KIF18A function in fertility and mitotic progression. *Scientific Reports.* 15:23046 PMID: PMC12214974
30. Semic A, Chan KY, Bernardi P, Silveria KC, Silveria C, Cavalcanti DP, Kannu P, Stumpff J. (2026) Novel KIF22 variants disrupt mitosis in human chondrocytes and expand SEMDJL2 mechanisms. *MBoC* 37(9): PMID: PMC13525594
31. Semic A*, Reddy BJN*, Muretta JM*, Vandal SE, Thompson AF, Fonseca CL, Rosenfeld SS, Gross SP, Stumpff J. (2026) A phosphorylation switch governs KIF11's mechanical output during mitosis. *PNAS* 123(33): PMID: PMC13486609

Preprints/ in Review

1. Fonseca C, Fisher K, Wagner E, Paschall SC, Kim H, Stumpff J. (2026) KIF18A maintains kinetochore-microtubule attachments in CIN cells by limiting microtubule polymerization. *bioRxiv*. DOI: 10.64898/2026.04.15.718686 (in revision)
2. Paschall SC, Blasius TL, Missman A, Rodriguez P, Cianfrocco MA, Verhey KJ, Stumpff J. A systems-level proteomic analysis identifies kinesin targets of KIFBP during neuronal development. *bioRxiv*. DOI: 10.64898/2026.07.17.739260. (in revision)

Non-Peer Reviewed Publications

Review Articles–

1. O'Farrell, P, Stumpff, J, and Su, TT 2004. Embryonic Cleavage Cycles: How is a mouse like a fly? *Current Biology* 14: R35-R45 (2004).
2. Su, TT and Stumpff, J (2004). Promiscuity Rules? The dispensability of Cyclin E and Cdk2. *Science STKE* 224: pe11
3. Stumpff, J and Wordeman, L (2007). Chromosome Congression: The Kinesin-8 step path to alignment. *Current Biology* 17, R326-328.

4. Stumpff, J and Asbury, C (2008). Chromosome bi-orientation: Euclidian euploidy. *Current Biology* 18, R81-R83.
5. Wordeman, L and Stumpff, J (2009). Microtubule Length Control, a Team Sport? *Developmental Cell* 17, 437-438
6. Stumpff J (2012) Measuring microtubule thickness: an exercise in cooperativity. *Dev Cell* 23: 1-2. PMID: 22814594.
7. Stumpff J, Ghule PN, Shimamura A, Stein JL, and Greenblatt M (2014) Spindle microtubule dysfunction and cancer predisposition. *J Cell Physiol.* 229:1881-1883. PMCID: PMC4149604.
8. Bissonette S and Stumpff J (2014). Quantifying mitotic chromosome dynamics and positioning. *J Cell Physiol* 229:1301-1305, PMID: 24683081. (featured on journal **cover**)
9. Malaby HLH and Stumpff J (2014). Microtubule recognition: A curvy attraction. *Curr Biol* 24: R998-R1000. PMID: 25442855.
10. Semic A and Stumpff J. (2026) Meiotic spindle mechanics: Sensing a disturbance in the force. *Current Biology* 36: R140-R143. PMID: 41734748 (Invited Review Article).

Books and Chapters—

1. J. Stumpff, J. Cooper, S. Domnitz, K. Rankin, M. Wagenbach, L. Wordeman. (2007). *In vitro and in vivo analysis of microtubule destabilizing kinesins* In A.O. Sperry (Ed.), Methods in Molecular Biology: Molecular Motors. Totowa, NJ: Humana Press Inc.
2. Fonseca C and Stumpff J (2016). *Quantification of mitotic chromosome alignment*. In Chang P and Ohi R (Ed.) The Mitotic Spindle: Methods and Protocols. Totowa, NJ: Humana Press Inc.
3. Thompson AF, Vandal S, Stumpff J. (2022). *Quantifying changes in chromosome position to access chromokinesin activity*. In Hinchcliffe EH (Ed.), Mitosis: Methods and Protocols. New York, NY: Springer Science + Business Media, LLC.

Other Scholarly Publications—

Abstracts

1. Semic A* and Stumpff J. Chondrocyte spindle biophysics in skeletal dysplasia. **2026** Mitotic Spindle: From Living and Synthetic Systems to Theory, Dubrovnik, Croatia.
2. Maredia Ebrahimi N and Stumpff J. KIF18A mutations lead to cell division errors: Insight into diminished fertility in young women. **2026** Pacific Coast Reproductive Society Meeting, Rancho Mirage, CA.
3. Semic A* and Stumpff J. Investigating the contributions of chondrocyte spindle biophysics and tissue microenvironment to skeletal dysplasia. **2025** ASCB Annual Meeting, Philadelphia, PA.
4. Paschall SC* and Stumpff J. Investigating mechanisms for spatial and temporal control of kinesins. **2024** ASCB Annual Meeting, San Diego, CA.
5. Semic A* and Stumpff J. Investigating the effects of the microenvironment on cell division. **2024** ASCB Annual Meeting, San Diego, CA.

6. Paschall SC* and Stumpff J. Investigating mechanisms for spatial and temporal control of kinesins. **2024** Cytoskeletal Motors Gordon Research Conference, Portland, ME.
7. Semic A* and Stumpff J. Investigating the effects of the microenvironment on cell division. **2024** Cytoskeletal Motors Gordon Research Conference, Portland, ME.
8. Queen KA*, Vandal SE, Thompson AF, and Stumpff J. KIF18A's neck-linker serves as a regulatable element to control microtubule subpopulation preference. **2023** Northeast Motile and Contractile Systems Conference, Burlington, VT.
9. Paschall SC* and Stumpff J. Exploring the role of kinesin family binding protein in mitosis and neuron development. **2023** Northeast Motile and Contractile Systems Conference, Burlington, VT.
10. Queen KA*, Vandal SE, Thompson AF, and Stumpff J. KIF18A's neck-linker serves as a regulatable element to control microtubule subpopulation preference. **2022** American Society for Cell Biology & European Molecular Biology Organization annual meeting, Washington, DC.
11. Queen KA*, Vandal SE, Thompson AF, and Stumpff J. KIF18A's neck-linker serves as a regulatable element to control microtubule subpopulation preference. **2022** Cytoskeletal Motors Gordon Research Conference, Mt. Snow, VT.
12. Paschall SC*, Schutt K, and Stumpff J. Exploring the role of kinesin binding protein in mitosis and neuron development. **2022** Cytoskeletal Motors Gordon Research Conference, Mt. Snow, VT.
13. Queen KA*, Vandal SE, Thompson AF, and Stumpff J. Phosphorylation of KIF18A alters its mitotic function and localization. **2021** Mitotic Spindle from Living and Synthetic Systems to Theory, Virtual.
14. Queen KA*, Vandal SE, Thompson AF, and Stumpff J. Phosphorylation of KIF18A alters its mitotic function and localization. **2021** Biophysical Society Annual Meeting, Virtual.
15. Thompson, A.F.*, Blackburn, P.R., Wagenbach, M., Wordeman, L., Lian, J.B., Klee, E.W., and Stumpff, J.K. Pathogenic mutations in the chromokinesin KIF22 disrupt anaphase chromosome segregation and cause skeletal dysplasia. **2020**. American Society for Cell Biology & European Molecular Biology Organization annual meeting, Virtual.
16. Sepaniac, L.,* Martin, W., Reinholdt, L.G., and Stumpff, J. Micronuclei arising due to loss of KIF18A form stable micronuclear envelopes and do not promote tumorigenesis. **2020** American Society for Cell Biology & European Molecular Biology Organization annual meeting, Virtual.
17. Queen KA*, Vandal SE, Thompson AF, and Stumpff J. Phosphorylation of KIF18A alters its mitotic function and localization. **2020** American Society for Cell Biology & European Molecular Biology Organization annual meeting, Virtual.
18. Thompson, A.F.*, Blackburn, P.R., Wagenbach, M., Wordeman, L., Lian, J.B., Klee, E.W., and Stumpff, J.K. Mutations in the chromokinesin KIF22 disrupt mitotic chromosome segregation and cause skeletal dysplasia. **2019**. American Society for Cell Biology & European Molecular Biology Organization annual meeting, Washington, DC.
19. Sepaniac, L.,* Martin, W., Reinholdt, L.G., and Stumpff, J. Micronuclei resulting from different insults differentially impact genomic stability. **2019** American Society for Cell Biology & European Molecular Biology Organization annual meeting, Washington, DC.

20. Queen KA*, Fonseca CL, Kim H, and Stumpff J. Investigating the Role of KIF18A in Satisfying the Spindle Assembly Checkpoint. **2019** American Society for Cell Biology & European Molecular Biology Organization annual meeting, Washington, D.C.
21. Thompson, A.F.*, Blackburn, P.R., Wagenbach, M., Wordeman, L., Lian, J.B., Klee, E.W., and Stumpff, J.K. Mutations in the motor and coiled-coil domains of the chromokinesin KIF22 cause skeletal dysplasia. **2019**. New England Genome Instability and Cancer Conference, Burlington, VT.
22. Thompson, A.F.*, Blackburn, P.R., Wagenbach, M., Wordeman, L., Lian, J.B., Klee, E.W., and Stumpff, J.K. Mutations in the chromokinesin KIF22 disrupt mitotic chromosome segregation and cause skeletal dysplasia. **2019**. Motile and Contractile Systems Gordon Research Conference, Colby-Sawyer College, NH.
23. Sepaniac, L.,* Martin, W., Reinholdt, L.G., and Stumpff, J. Micronuclei resulting from different insults, differentially impact genomic stability. **2019** EMBO Chromosome Segregation and Aneuploidy Workshop, Cascais, Portugal.
24. Malaby, H.L.H.*, Lessard, D.V., Berger, C.L., and Stumpff, J. Where the K-fiber ends: The importance of Kif18A neck linker length for mitotic localization. **2018**. Gordon Research Conference Cytoskeletal Motors, Mt. Snow, VT.
25. Thompson, A.F.*, Vandal, S.E., Blackburn, P.R., Wagenbach, M., Wordeman, L., Lian, J.B., Klee, E.W. and Stumpff, J.K. **2018**. Mutations in the motor and coiled-coil domains of the chromokinesin KIF22 cause skeletal dysplasia. **2018**. Cytoskeletal Motors Gordon Research Conference, West Dover, VT.
26. Sepaniac, L.,* Reinholdt, L.G., and Stumpff, J. Different mechanisms of micronucleus formation and impact to genomic stability. **2018**. New England Genome Instability and Cancer Conference, Worcester Polytechnic Institute, Worcester, MA.
27. Thompson, A.F.*, Muretta, J.M., Reddy, B.J.N., Scarabelli, G., Jariwala, S., Major, J., Venere, M., Rich, J.N., Willard, B., Thomas, D.D., Stumpff, J.K., Grant, B.J., Gross, S.P. and Rosenfeld, S.S. Effects of post-translational modification on the mitotic kinesin Eg5. **2018**. New England Genome Instability and Cancer Conference, Worcester, MA.
28. Thompson, A.F.*, Muretta, J.M., Reddy, B.J.N., Scarabelli, G., Jariwala, S., Major, J., Venere, M., Rich, J.N., Willard, B., Thomas, D.D., Stumpff, J.K., Grant, B.J., Gross, S.P. and Rosenfeld, S.S. Effects of post-translational modification on the mitotic kinesin Eg5. **2018**. Boston Area Mitosis and Meiosis Meeting, Cambridge, MA.
29. Sepaniac, L.,* Reinholdt, L.G., and Stumpff, J. Different mechanisms of micronucleus formation and impact to genomic stability. **2018**. University of Vermont Cancer Center Clinical and Translational Research Symposium, University of Vermont Medical Center, Burlington, VT.
30. Sepaniac, L.,* Reinholdt, L.G., and Stumpff, J. Different mechanisms of micronucleus formation and impact to genomic stability. **2018**. National Council of Space Grant Director's Spring Meeting, Crystal City, VA.
31. Clayton, J.*, Fonseca, C., Marquis, C. Wood, L., Anker, C., Howe, A., and Stumpff, J. Investigating mitotic kinesins as therapeutic targets for triple negative breast cancer. **2017**. American Society for Cell Biology (ASCB) Annual Meeting, Philadelphia, PA.

32. Malaby, H.L.H.*, Dumas, M., Ohi, R., and Stumpff, J. Kinesin binding protein is an inhibitory regulator of the mitotic kinesins Kif18A and Kif15. **2017**. American Society for Cell Biology (ASCB) Annual Meeting, Philadelphia, PA.
33. Thompson, A.F.*, Muretta, J.M., Reddy, B.J.N., Scarabelli, G., Jariwala, S., Major, J., Venere, M., Rich, J.N., Willard, B., Thomas, D.D., Stumpff, J.K., Grant, B.J., Gross, S.P. and Rosenfeld, S.S. Effects of post-translational modification on the mitotic kinesin Eg5. **2017**. American Society for Cell Biology/European Molecular Biology Organization Annual Meeting, Philadelphia, PA.
34. Sepaniac, L.,* Reinholdt, L.G., and Stumpff, J. Different mechanisms of micronucleus formation and impact to genomic stability. **2017**. American Society for Cell Biology Annual Conference, Philadelphia, PA.
35. Marquis, C., Fonseca, C., Wood, L., Anker, C., Wood, M., Clayton, J., Stumpff, J. Exploring New Therapeutic Strategies for Triple Negative Breast Cancer. **2017**. Women's Health and Cancer Conference, Sheraton Burlington Hotel & Conference Center, VT.
36. Sepaniac, L.,* Reinholdt, L.G., and Stumpff, J. Different mechanisms of micronucleus formation and impact to genomic stability. **2017**. Human and Mammalian Genetics and Genomics Symposium: 58th Annual McKusick Short Course, The Jackson Laboratory, Bar Harbor, ME.
37. Kim, H.* and Stumpff, J. Kif18A coordinates the alignment and attachment of chromosomes during cell division. **2017**. Motile and Contractile Systems, Gordon Research Conference, Colby-Sawyer College, NH.
38. Martin, W., Thompson, A.F.*, Byers, C., Stumpff, J.K. and Reinholdt, L.G. A potential role for the kinetochore protein kinetochore-associated 1 (KNTC1) in ciliary signaling. **2017**. Motile and Contractile Systems Gordon Research Conference, Colby-Sawyer College, NH.
39. Sepaniac, L.,* Reinholdt, L.G., and Stumpff, J. Different mechanisms of micronucleus formation and impact to genomic stability. **2017**. LXXXII Cold Spring Harbor Symposium: Chromosome Segregation and Structure, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY.
40. Malaby, H.L.H.* and Stumpff, J. Where the K-fiber ends: Elucidating the mechanism of Kif18A localization in mitosis. **2017**. Boston Area Mitosis and Meiosis (BAMM) Annual Meeting, Whitehead Institute, Boston, MA.
41. Sepaniac, L.,* Reinholdt, L.G., and Stumpff, J. Different mechanisms of micronucleus formation and impact to genomic stability. **2017**. Boston Area Mitosis and Meiosis Meeting, Whitehead Institute for Biomedical Research, Cambridge, MA.
42. Sepaniac, L.,* Fonseca, C. and Stumpff, J. Centering the Genome: Why do chromosomes line up at the spindle equator? **2017**. Chromosome and Cytoskeleton Dynamics Conference, University of Massachusetts, Amherst, MA.
43. Kim, H.* and Stumpff, J. Kif18A coordinates chromosome alignment and attachment of chromosomes during cell division. **2016**. American Society for Cell Biology Annual meeting, San Francisco, CA. (Oral presentation)
44. Sepaniac, L.,* Reinholdt, L.G., and Stumpff, J. Different mechanisms of micronucleus formation and impact to genomic stability. **2016**. New England Nuclear Envelope Meeting, Yale University, New Haven, CT.

45. Malaby, H.* and Stumpff, J. Where the K-fiber ends: Elucidating the mechanisms that spatially confine chromosomes in mitosis **2016**. Molecular Motors Gordon Conference, Mt. Snow, VT
46. Kim, H.*, Fonseca, C., Czechanski, A., Byers, C., Greenstein, I., Reinholdt, L., and Stumpff, J. Temporally coordinating chromosome alignment with stabilization of kinetochore-microtubule attachments. **2016**. Boston Area Mitosis and Meiosis meeting. Whitehead Institute, Boston, MA.
47. Malaby, H.* and Stumpff, J. Where the K-fiber ends: Elucidating the mechanisms that spatially confine chromosomes in mitosis **2016**. Boston Area Mitosis and Meiosis meeting. Whitehead Institute, Boston, MA.
48. Fonseca, C., Messinger, D., Tang, M., Reinholdt, L. and Stumpff, J.*. Centering the genome: Why do mitotic chromosomes line up at the spindle equator? **2015**. Chromosome Dynamics Gordon Research Conference, Waterville Valley, NH.
49. Malaby, H.* and Stumpff, J. Chromosome corralling: Elucidating the mechanisms that spatially confine chromosomes in mitosis. **2015**. Motile and Contractile Systems, Gordon Research Conference, Colby-Sawyer College, NH.
50. Kim, H.*, Fonseca C., Czechanski, A., Byers, C., Greenstein, I., Reinholdt, LG. and Stumpff, J. Kinesin-like motors Kif18A and Kid modulate kinetochore-microtubule attachments. **2015**. Boston Area Mitosis and Meiosis meeting, Boston, MA.
51. Fonseca, C.*, Messinger, D., Tang, M., Reinholdt, L. and Stumpff, J. Mitotic chromosome alignment is dispensable for accurate segregation. **2015**. Boston Area Mitosis and Meiosis meeting, Boston, MA.
52. Malaby, H.* and Stumpff, J. Chromosome corralling: Elucidating the mechanisms that spatially confine chromosomes in mitosis. **2015**. Boston Area Mitosis and Meiosis meeting, Boston, MA.
53. Fonseca, C., Messinger, D., Tang, M., Reinholdt, L. and Stumpff, J.* Mitotic chromosome alignment is dispensable for accurate segregation. **2014**. ASCB annual meeting. Philadelphia, PA.
54. Kim, H.*, Fonseca, C., Czechanski, A., Byers, C., Greenstein, I., Reinholdt, LG. and Stumpff, J. Kif18A stabilizes kinetochore-microtubule attachments. **2014**. ASCB annual meeting. Philadelphia, PA.
55. Fonseca, C.* and Stumpff, J. Chromosome alignment is not required for accurate segregation. **2014**. Boston Area Mitosis and Meiosis meeting, Boston, MA.
56. Kim, H.*, Greenstein, I., Reinholdt, LG. and Stumpff, J. Mutations in R308 of Kif18A cause primordial germ cell depletion in mice. **2014**. Boston Area Mitosis and Meiosis meeting, Boston, MA.
57. Kim, H., Fonseca, C. and Stumpff, J.* Molecular control of microtubule length within the spindle. **2013**. ASCB annual meeting, New Orleans, LA.
58. Kim, H., Fonseca, C. and Stumpff, J.* Molecular control of kinetochore-microtubule length. **2013**. Motile and Contractile Systems, Gordon Research Conference, Colby-Sawyer College, NH.

59. Kim, H.*, Greenstein, I., Reinholdt, LG. and Stumpff, J. Mutations in R308 of Kif18A cause primordial germ cell depletion in mice. **2013**. Motile and Contractile Systems, Gordon Research Conference, Colby-Sawyer College, NH.
60. Fonseca, C. and Stumpff, J.* Molecular control of spindle microtubule lengths. **2012**. ASCB annual meeting, San Francisco, CA.
61. Stumpff, J.*, Fonseca, C., Thompson, J., M. Wagenbach and Wordeman, L. Mitotic chromosome alignment is antagonistically controlled by two microtubule growth suppressing kinesins. **2012**. FASEB Mitosis and Spindle Function Conference, Steamboat Springs, CO.
62. Du, Y.*, Stumpff, J.*, English, C., Maliga, Z., Wagenbach, M., Asbury, C., Wordeman, L. and Ohi, R. A tethering mechanism controls the processivity and kinetochore-microtubule plus-end enrichment of the kinesin-8 Kif18A. **2011**. ASCB annual meeting, Denver, CO.
63. Stumpff, J.*, M. Wagenbach, G. Franck, A., Huseby, C., Asbury, C. and Wordeman, L. Kif18A and chromokinesins use microtubule growth suppression and spatial control of kinetochore tension to align chromosomes. **2011**. Motile and Contractile Systems, Gordon Research Conference, Colby-Sawyer College, NH.

* indicates primary presenter

Patents Issued or Pending-

U.S. Provisional Patent Application Serial No. 63/022,885, filed May 11, 2020

A Treatment Approach Involving KIF18A Inhibition for Chromosomally Unstable Tumors

Inventor: Jason Stumpff

PCT Application No. PCT/US2021/031761, filed May 11, 2021

A Treatment Approach Involving KIF18A Inhibition for Chromosomally Unstable Tumors

Inventor: Jason Stumpff

SUMMARY OF SCHOLARLY ACTIVITIES

My laboratory uses advanced microscopy techniques to investigate the mechanisms that control cytoskeletal and chromosome dynamics during cell division and differentiation. Abnormal control of chromosome movements leads to chromosome segregation errors and the production of aneuploid cells, i.e. cells with the wrong number of chromosomes. Aneuploidy is implicated in the initiation and development of cancer, is a leading cause of human miscarriages, and underlies monosomy and trisomy syndromes (e.g. Turner's syndrome, Down's syndrome, Edward's syndrome and Patau syndrome). Thus, elucidating the mechanisms that control chromosome movements during cell division is an important step towards understanding the molecular basis of a wide range of human health disorders and identifying novel treatment targets.

Recent projects have focused on understanding how the structures of molecular motors are tuned for specific cellular functions and in determining the basis of cell and tissue specific requirements for motor proteins that have direct links to human disease. These studies have led to identification of disease mechanisms and therapeutic targets. For example, our body of work on the mitotic motor protein KIF18A has contributed to development of anti-KIF18A drugs that are being clinically tested as treatments for aggressive cancers.

INVITED PRESENTATIONS**Regional Conferences**

1. 2019 Northeast Genomic Instability Conference, UVM, Burlington, VT
2. 2018 Women's Health and Cancer Conference, Burlington, VT (keynote speaker)
3. 2017 Women's Health and Cancer Conference, Burlington, VT
4. 2017 Northeast Genomic Instability Conference, Dartmouth College, Hanover, NH
5. 2017 ASCB Chromosome and Cytoskeletal Dynamics Meeting, UMass, Amherst, MA
6. 2016 Northeast Genomic Instability Conference, University of Vermont, Burlington, VT
7. 2016 Boston Area Mitosis and Meiosis Meeting, Whitehead Institute, Boston, MA
8. 2015 NE Regional Chromosome Pairing Conf, Harvard Medical School, Boston, MA
9. 2014 NE Regional Chromosome Pairing Conf, Harvard Medical School, Boston, MA
10. 2014 Boston Area Mitosis and Meiosis Meeting, Whitehead Institute, Boston, MA
11. 2013 NE Regional Chromosome Pairing Conference, Dartmouth College, Hanover, NH
12. 2012 Leukemia and Lymphoma Society Special Presentation, Poughkeepsie, NY
13. 2012 Chromosome Pairing Conference, Dartmouth University, Hanover, NH
14. 2012 Northeast Genomic Instability Conference, University of Vermont, Burlington, VT
15. 2011 Clinical and Translational Research Symp, University of Vermont, Burlington, VT

Departmental Seminars

1. 2025 Radiobiology and Genome Integrity Program, Yale University, New Haven, CT
2. 2024 Genetics, Rutgers University, New Brunswick, NJ
3. 2024 Cell and Developmental Biology, Univ of Colorado Anschutz, Denver, CO
4. 2024 Biomedical and Molecular Sciences, Queens University, Kingston, ON
5. 2023 Redox Biology Seminar Series, Univ of Vermont, Burlington, VT
6. 2023 Cell Biology and Anatomy, McGill University, Montreal, Canada
7. 2023 Institute for Research in Cancer and Immunology, Montreal, Canada
8. 2022 Medicine Grand Rounds, Univ of Vermont, Burlington, VT
9. 2022 Cell and Developmental Biology, Univ of Michigan, Ann Arbor, MI
10. 2022 Hematology and Oncology Grand Rounds, Univ of Vermont, Burlington, VT
11. 2022 Cell and Developmental Biology, Northwestern University, Chicago, IL
12. 2021 Department of Biology, University of Konstanz, Konstanz, Germany
13. 2021 Virginia Tech Life Science Seminar, Virginia Tech, Blacksburg, VA
14. 2021 Firestone Medical Research Building Campaign Committee, Burlington, VT
15. 2020 Dept of Molecular, Cellular, and Developmental Biology, Univ of CO, Boulder, CO
16. 2019 Dept of Microbiology and Molecular Genetics, Univ of Vermont, Burlington, VT
17. 2019 Department of Plant Biology, University of Vermont, Burlington, VT
18. 2019 Department of Biology, University of Vermont, Burlington, VT
19. 2018 Department of Molecular Biology, University of Wyoming, Laramie, WY
20. 2017 UVM Cancer Center Advisory Board, University of Vermont, Burlington, VT
21. 2017 OBGYN Annual Retreat, University of Vermont, Burlington, VT
22. 2017 Department of Biological Sciences, Rensselaer Polytechnic Institute, Troy, NY
23. 2017 Dept of Biology and Biotechnology, Worcester Polytechnic Inst, Worcester, MA
24. 2016 Dept of Cell and Developmental Biology, University of Colorado, Denver, CO
25. 2016 Department of Biochemistry, Colorado State University, Fort Collins, CO
26. 2015 Dept of Biochemistry, University of Vermont, Burlington, VT
27. 2014 OBGYN Annual Retreat, University of Vermont, Burlington, VT
28. 2014 Department of Biochemistry Seminar, Dartmouth College, Hanover, NH

29. 2014 Vermont Cancer Center Seminar, University of Vermont, Burlington, VT
30. 2013 Department of Biology, University of Vermont, Burlington, VT
31. 2013 The Jackson Laboratory Seminar Series, The Jackson Laboratory, Bar Harbor, ME
32. 2013 Department of Biology, Bowdoin College, New Brunswick, ME
33. 2012 Vermont Cancer Center Grand Rounds, University of Vermont, Burlington, VT
34. 2012 Environmental Pathology Seminar Series, University of Vermont, Burlington, VT
35. 2012 Dept of Microbiology and Molecular Genetics, Univ of Vermont, Burlington, VT
36. 2011 Department of Pharmacology, University of Vermont, Burlington, VT
37. 2010 Molecular Physiology and Biophysics Dept, Univ of Vermont, Burlington, VT
38. 2009 Patton Symposium, University of Washington
39. 2004 Department of Embryology, Carnegie Institute of Washington
40. 2004 Department of Physiology and Biophysics, University of Washington
41. 2004 Department of Cellular and Molecular Medicine, UCSD

International Conferences

1. 2023 Mitotic Spindle: From Living and Sythetic Systems to Theory, Dubrovnik, Croatia
2. 2022 FASEB Consequences of Aneuploidy Meeting, Southbridge, MA
3. 2019 American Society for Cell Biology, Washington DC
4. 2019 Mitotic Spindle: From Living and Synthetic Systems to Theory, Split, Croatia
5. 2017 American Society for Cell Biology, Philadelphia, PA
6. 2016 Muscle and Molecular Motors Gordon Conference, Mt. Snow, VT
7. 2015 Dynamic Kinetochore Workshop, Copenhagen, Denmark
8. 2014 American Society for Cell Biology Annual Meeting, Philadelphia, PA
9. 2014 Molecular Motors Gordon Research Conference, Mt. Snow, VT
10. 2013 American Society for Cell Biology Annual Meeting, New Orleans, MS
11. 2013 The Dynamic Kinetochore Workshop. Porto, Portugal
12. 2007 FASEB Mitosis and Spindle Function meeting